Real Madrid ACWR Project

Student Presentation Handout

One-Sentence Summary

This project is an interactive tool that helps football fitness coaches test a future training plan and see whether each player's workload is likely to become too low, normal, or risky over the next 15 days.

1. The Main Sport-Science Idea

The project focuses on ACWR, which means Acute:Chronic Workload Ratio.

- Acute workload: how much load the player has had recently, roughly the last 7 days.
- Chronic workload: the player's longer-term fitness/load base, roughly the last 28 days.
- ACWR: acute workload divided by chronic workload.

In simple terms: ACWR = recent workload / normal longer-term workload.

If a player suddenly has a very hard week compared with what he has been used to over the last month, his ACWR rises.

2. How To Interpret ACWR

Undertraining: ACWR < 0.8. The player may not be receiving enough load stimulus.

Optimal: ACWR 0.8 to 1.3. The player is in a generally safe training range.

Caution: ACWR 1.3 to 1.5. The player may be approaching elevated workload risk.

Danger: ACWR >= 1.5. The player may be overloaded and should be monitored closely.

Important: this tool does not predict injuries directly. It predicts workload risk signals. Coaches should combine it with medical information, wellness data, player feedback, tactical needs, and staff judgment.

3. What The Tool Does

The tool lets coaches answer questions like: If we plan these sessions for the next 15 days, which players might enter a risky workload zone?

The coach does not need to write code. They use a Streamlit app with two main parts:

- Dashboard: shows the current ACWR status of the squad.
- Planning and Forecast: lets the coach create a 15-day plan and then forecasts future ACWR.

Planned session types: G = game-based work / small-sided games; TAC = tactical; BP = set pieces; TEC = technical; MATCH = official match; REST = no planned load.

4. The Three Workload Metrics

The project calculates ACWR separately for three football GPS/IMU metrics:

- Total Distance: overall running volume. This reflects general aerobic and movement load.
- Accelerations: high-intensity acceleration efforts. This reflects neuromuscular stress.
- High-Speed Running: distance covered at high speed. This is important for sprint exposure and high-speed running demands.

This matters because a player can look safe in total distance but risky in high-speed running, or the opposite.

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5. How The Project Works

1. Historical player data is collected: past sessions, matches, player positions, height, weight, age, and GPS workload metrics.
2. The data is cleaned and organized by day: the original drill-level data becomes one row per player per day.
3. The project learns patterns from the past: for example, how much load a tactical session usually creates for a player or position.
4. The coach enters a future plan: for example, tactical session tomorrow, match in three days, recovery day after that.
5. The model predicts future daily loads for every player and every future day.
6. The app recalculates future ACWR by combining each player’s real past workload with the predicted future workload.
7. The app shows risk zones: optimal, undertraining, caution, or danger.

6. Simple Analogy

Think of each player like a car engine.

- The chronic workload is what the engine is used to handling.
- The acute workload is what we are asking it to do this week.
- If we suddenly ask too much compared with its normal level, the risk light turns on.

7. Suggested Presentation Structure

1. Problem: coaches need to plan training while avoiding sudden workload spikes.
2. Sport-Science Concept: explain acute workload, chronic workload, ACWR, and the risk zones.
3. Data: football training and match data, player profile, position, session type, and GPS load metrics.
4. Model: the project learns from previous sessions to estimate future load.
5. App: a coach creates a 15-day schedule, then the app forecasts ACWR and highlights risk alerts.
6. Value: faster planning, player-specific monitoring, and clearer communication between data staff and fitness coaches.
7. Limitations: no wellness or RPE data, ACWR is not a medical diagnosis, and results depend on historical data quality.

8. One-Minute Presentation Script

This project is a workload planning tool for football fitness coaches. It uses past GPS training and match data to understand how different session types affect each player. The key metric is ACWR, which compares recent workload over about 7 days with longer-term workload over about 28 days. If the recent load is too high compared with the player’s normal base, the player may enter a caution or danger zone.

In the app, a coach builds a 15-day training plan, and the system predicts how each player’s total distance, accelerations, and high-speed running will evolve. Then it calculates future ACWR and highlights players who may be undertrained or overloaded. The goal is not to replace coaches, but to give them a clear decision-support tool before finalizing the training plan.

9. Short Version For A Slide

Project purpose: forecast player workload risk before a training plan is finalized.

Input: past GPS data + player profile + planned 15-day session schedule.

Output: future ACWR curves and risk zones for each player.

Main value: helps coaches identify players who may be undertrained or overloaded.